

Gaurika Mahajan

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EDUCATION

University of Toronto

Toronto, ON | Sept 2022 - May 2027

BASc in Computer Engineering, Minor in Artificial Intelligence

Relevant Courses: Data Structures and Algorithms, Operating Systems, Computer Networks, Databases, Computer Security, Deep Learning

TECHNICAL SKILLS

Languages and Frameworks: Python, Golang, SQL, PySpark, Java, C++ , C

Tools: Databricks, Apache Kafka, Airflow, Docker, Kubernetes, Terraform, Jenkins, Datadog, Splunk

Databases/Cloud: AWS (S3, EC2, EKS, Lambda, DynamoDB), Elasticsearch, PostgreSQL, MySQL, MongoDB, Redis

WORK EXPERIENCE

Software Engineering Intern | Patreon

New York, NY | May 2026 - Aug 2026

- Architected a diff-based framework to update database rows upon change instead of full rewrites, reducing redundant writes by **80%** and runtime by **20%** across **20 production pipelines**.
- Designed **Delta Lake** tables mirroring Elasticsearch indices to pinpoint which rows change every run, at **300M+ row scale**.
- Introduced a concurrency-safe write, adopted framework-wide, eliminating **7+ weekly** race-condition failures.
- Built automated reconciliation between Databricks and Elasticsearch catching **67K+ missing records**, and added GDPR-compliant deletion, keeping user-facing data complete, accurate, and compliant.

Software Engineering Intern | Zynga

Toronto, ON | May 2025 – April 2026

- Enhanced Go-based Kafka consumer handling **500K+** daily events, resolving schema error codes to reduce message failures.
- Strengthened CI/CD infrastructure by integrating SonarQube across **8+ repositories**, reducing code redundancy by **65%**.
- Integrated RPM hosting infrastructure with automated deployment via **Jenkins CI** to DNF server, increasing reliability by **28%**.
- Extended player-level **Splunk** logging controls to event streaming system by implementing Go and Redis filtering across admin tool and event consumer, ensuring user privacy compliance while optimizing log volume for 10+ games.

Full Stack Web Developer | UofT Engineering Orientation Website

Remote | May 2024 – Aug 2024

- Built and deployed a full-stack **React/Redux** and **Express** app serving **1,000+ students**, processing **\$150K+** via **Stripe**.
- Optimized backend with **MongoDB** indexing for **40%** faster queries and **Redis** caching for **20% higher scalability**.

Compliance Intern | Fresenius Kabi USA

Chicago, IL | May 2024 – Aug 2024

- Designed monorepo architecture consolidating 7 repositories into single workflow, **improving data retrieval speed by 95%**.
- Analyzed **100k+ datapoints across 12 years** to design a database schema, cutting launch forecasting **from 6 hours to seconds**.

LEADERSHIP

Software Development Director | UofT You're Next Career Network

April 2024 – Present

- Led team of 10 developers in expanding **React/Firebase** platform serving **3,500+ students** to manage event sign-ups.
- Defined product roadmap, created technical specs for feature development, and designed user-centric interface in **Figma**.

Web Development Director | UofT Women in Science & Engineering

Aug 2023 – Aug 2024

- Led website development and infrastructure improvements, enhancing user experience for **500+ members**.
- Architected automated email management system using **Gmail APIs**, reducing manual administrative overhead by **60%**.

PROJECTS

GIS Mapping Software | C++ , Git, Valgrind

- Built and optimized a GIS application in a team of 3 using an **Agile-inspired approach** and **OpenStreetMap API**
- Implemented **A* and Dijkstra algorithms** for efficient pathfinding, leveraging **multi-threading** to cut latency by **200%**

Code Canvas | Docker, LLM Integration, C4 Modeling, Claude Sonnet 4, Prompt Engineering

- Built AI-powered system architecture diagram generator via **Claude Sonnet 4** and C4 modeling to analyze repos **in minutes**.
- Engineered containerized solution capturing **85%** of codebase architecture, to produce system, infra, and component diagrams.
- Eliminated outdated documentation across **15+ engineering teams**, reducing onboarding time by **60%**.

Skin Cancer Classifier | Python, PyTorch, TensorFlow, Google Colab, Git

- Developed a AI-driven CNN-based skin cancer classifier, achieving **80% test accuracy** on dermatoscopic images.
- Integrated unsupervised learning techniques for data augmentation and feature extraction to enhance training data quality.
- Led data collection, model creation, training, and hyperparameter optimization for improved classification performance.